

MLR Part II

Multiple Linear Regression

The Boston data set contains 12 variables, and so it would be cumbersome to have to type all of these in order to perform a regression using all of the predictors. Instead, we can use the following short-hand:

```
library(MASS)
library(ISLR2)

##
## Attaching package: 'ISLR2'

## The following object is masked from 'package:MASS':
##
## Boston

lm.fit <- lm(medv ~ ., data = Boston)
summary(lm.fit)

##
## Call:
## lm(formula = medv ~ ., data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1304  -2.7673  -0.5814   1.9414  26.2526
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.617270   4.936039   8.431 3.79e-16 ***
## crim        -0.121389   0.033000  -3.678 0.000261 ***
## zn           0.046963   0.013879   3.384 0.000772 ***
## indus        0.013468   0.062145   0.217 0.828520
## chas         2.839993   0.870007   3.264 0.001173 **
## nox        -18.758022   3.851355  -4.870 1.50e-06 ***
## rm           3.658119   0.420246   8.705 < 2e-16 ***
## age          0.003611   0.013329   0.271 0.786595
## dis        -1.490754   0.201623  -7.394 6.17e-13 ***
## rad          0.289405   0.066908   4.325 1.84e-05 ***
## tax        -0.012682   0.003801  -3.337 0.000912 ***
## ptratio     -0.937533   0.132206  -7.091 4.63e-12 ***
## lstat       -0.552019   0.050659 -10.897 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.798 on 493 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7278
## F-statistic: 113.5 on 12 and 493 DF,  p-value: < 2.2e-16
```

We can access the individual components of a summary object by name (type `?summary.lm` to see what is

available). Hence `summary(lm.fit)$r.sq` gives us the R^2 , and `summary(lm.fit)$sigma` gives us the RSE.

General F test, Can we drop Age and Indus from the model?

$H_o: \beta_{Age} = \beta_{Indus} = 0$

H_a : One of the β is significant

Reduced model without Age and Indus

```
lm.red<- lm(medv ~ crim+zn+chas+nox+rm+dis+rad+tax+ptratio+lstat, data = Boston)
anova(lm.red,lm.fit)
```

```
## Analysis of Variance Table
##
## Model 1: medv ~ crim + zn + chas + nox + rm + dis + rad + tax + ptratio +
##      lstat
## Model 2: medv ~ crim + zn + indus + chas + nox + rm + age + dis + rad +
##      tax + ptratio + lstat
##      Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1      495 11352
## 2      493 11349   2    2.7677 0.0601 0.9417
```

Here Model 1 represents the reduced model, while Model 2 corresponds to full model. The `anova()` function performs a hypothesis test comparing the two models. P value is 0.94, we will accept H_o , we can drop Age and Indus.

Dropping variables

What if we would like to perform a regression using all of the significant variables?

For example, in the above regression output, `age` has a high p -value. So we may wish to run a regression excluding this predictor.

The following syntax results in a regression using all predictors except `age`.

```
lm.fit1 <- lm(medv ~ . - age, data = Boston)
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ . - age, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1851  -2.7330  -0.6116   1.8555  26.3838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  41.525128   4.919684   8.441 3.52e-16 ***
## crim        -0.121426   0.032969  -3.683 0.000256 ***
## zn           0.046512   0.013766   3.379 0.000785 ***
## indus        0.013451   0.062086   0.217 0.828577
## chas         2.852773   0.867912   3.287 0.001085 **
## nox        -18.485070   3.713714  -4.978 8.91e-07 ***
## rm           3.681070   0.411230   8.951 < 2e-16 ***
## dis        -1.506777   0.192570  -7.825 3.12e-14 ***
## rad           0.287940   0.066627   4.322 1.87e-05 ***
## tax        -0.012653   0.003796  -3.333 0.000923 ***
```

```
## ptratio      -0.934649    0.131653   -7.099 4.39e-12 ***
## lstat        -0.547409    0.047669  -11.483 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.794 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF,  p-value: < 2.2e-16
```

Alternatively, the `update()` function can be used.

```
lm.fit1 <- update(lm.fit, ~ . - indus )
summary(lm.fit1)
```

```
##
## Call:
## lm(formula = medv ~ crim + zn + chas + nox + rm + age + dis +
##      rad + tax + ptratio + lstat, data = Boston)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-15.1267	-2.7487	-0.5902	1.9056	26.2609

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	41.543721	4.919606	8.445	3.42e-16 ***
crim	-0.121628	0.032950	-3.691	0.000248 ***
zn	0.046642	0.013786	3.383	0.000773 ***
chas	2.859128	0.864680	3.307	0.001013 **
nox	-18.534872	3.707573	-4.999	8.01e-07 ***
rm	3.650015	0.418175	8.728	< 2e-16 ***
age	0.003608	0.013317	0.271	0.786563
dis	-1.499953	0.196913	-7.617	1.33e-13 ***
rad	0.285390	0.064231	4.443	1.09e-05 ***
tax	-0.012320	0.003411	-3.611	0.000336 ***
ptratio	-0.933839	0.130976	-7.130	3.59e-12 ***
lstat	-0.551115	0.050438	-10.927	< 2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.793 on 494 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7284
## F-statistic: 124.1 on 11 and 494 DF,  p-value: < 2.2e-16
```

Lets also get rid of indus variable that is also not significant

```
lm.fit2 <- update(lm.fit1, ~ . - age)
summary(lm.fit2)
```

```
##
## Call:
## lm(formula = medv ~ crim + zn + chas + nox + rm + dis + rad +
##      tax + ptratio + lstat, data = Boston)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
--	-----	----	--------	----	-----

```
## -15.1814 -2.7625 -0.6243 1.8448 26.3920
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41.451747  4.903283   8.454 3.18e-16 ***
## crim        -0.121665  0.032919  -3.696 0.000244 ***
## zn           0.046191  0.013673   3.378 0.000787 ***
## chas         2.871873  0.862591   3.329 0.000935 ***
## nox        -18.262427  3.565247  -5.122 4.33e-07 ***
## rm           3.672957  0.409127   8.978 < 2e-16 ***
## dis         -1.515951  0.187675  -8.078 5.08e-15 ***
## rad           0.283932  0.063945   4.440 1.11e-05 ***
## tax         -0.012292  0.003407  -3.608 0.000340 ***
## ptratio     -0.930961  0.130423  -7.138 3.39e-12 ***
## lstat       -0.546509  0.047442 -11.519 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.789 on 495 degrees of freedom
## Multiple R-squared:  0.7342, Adjusted R-squared:  0.7289
## F-statistic: 136.8 on 10 and 495 DF,  p-value: < 2.2e-16
```

#we can drop them at the same time

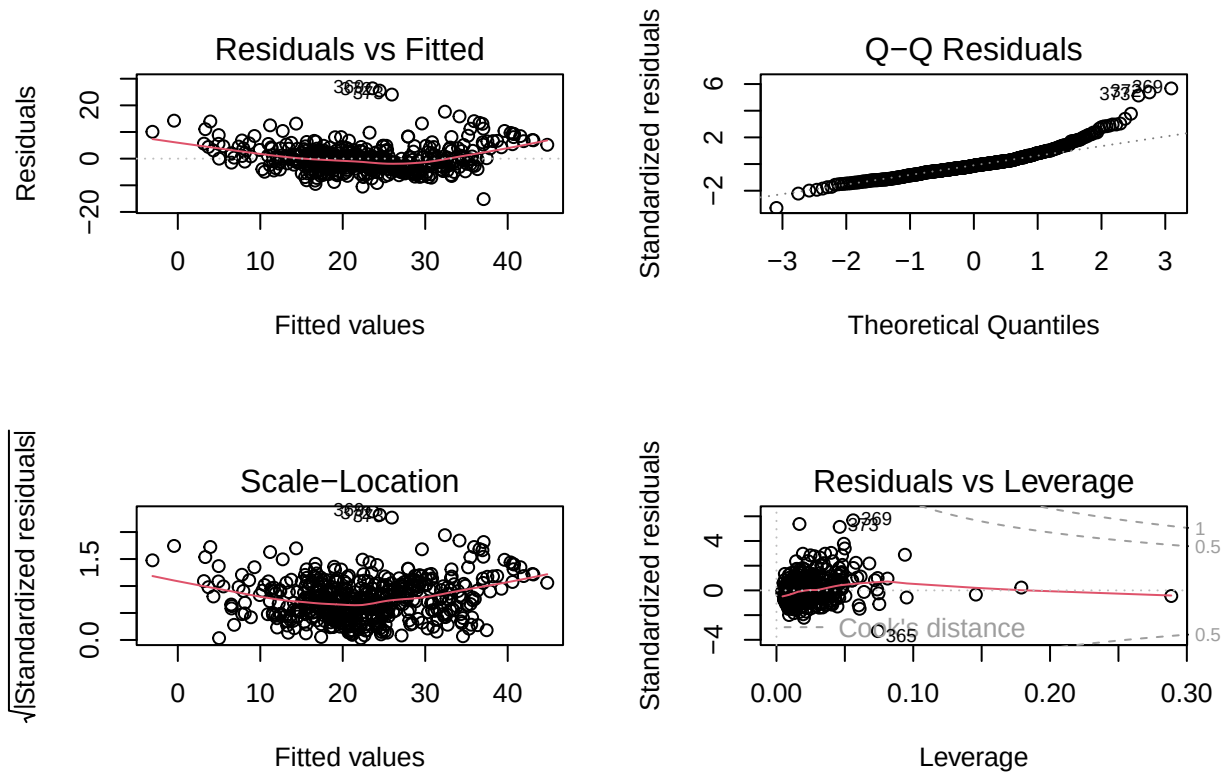
```
lm.fit2 <- update(lm.fit, ~ . - age - indus)
summary(lm.fit2)
```

```
##
## Call:
## lm(formula = medv ~ crim + zn + chas + nox + rm + dis + rad +
##      tax + ptratio + lstat, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1814  -2.7625  -0.6243   1.8448  26.3920
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 41.451747  4.903283   8.454 3.18e-16 ***
## crim        -0.121665  0.032919  -3.696 0.000244 ***
## zn           0.046191  0.013673   3.378 0.000787 ***
## chas         2.871873  0.862591   3.329 0.000935 ***
## nox        -18.262427  3.565247  -5.122 4.33e-07 ***
## rm           3.672957  0.409127   8.978 < 2e-16 ***
## dis         -1.515951  0.187675  -8.078 5.08e-15 ***
## rad           0.283932  0.063945   4.440 1.11e-05 ***
## tax         -0.012292  0.003407  -3.608 0.000340 ***
## ptratio     -0.930961  0.130423  -7.138 3.39e-12 ***
## lstat       -0.546509  0.047442 -11.519 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.789 on 495 degrees of freedom
## Multiple R-squared:  0.7342, Adjusted R-squared:  0.7289
## F-statistic: 136.8 on 10 and 495 DF,  p-value: < 2.2e-16
```

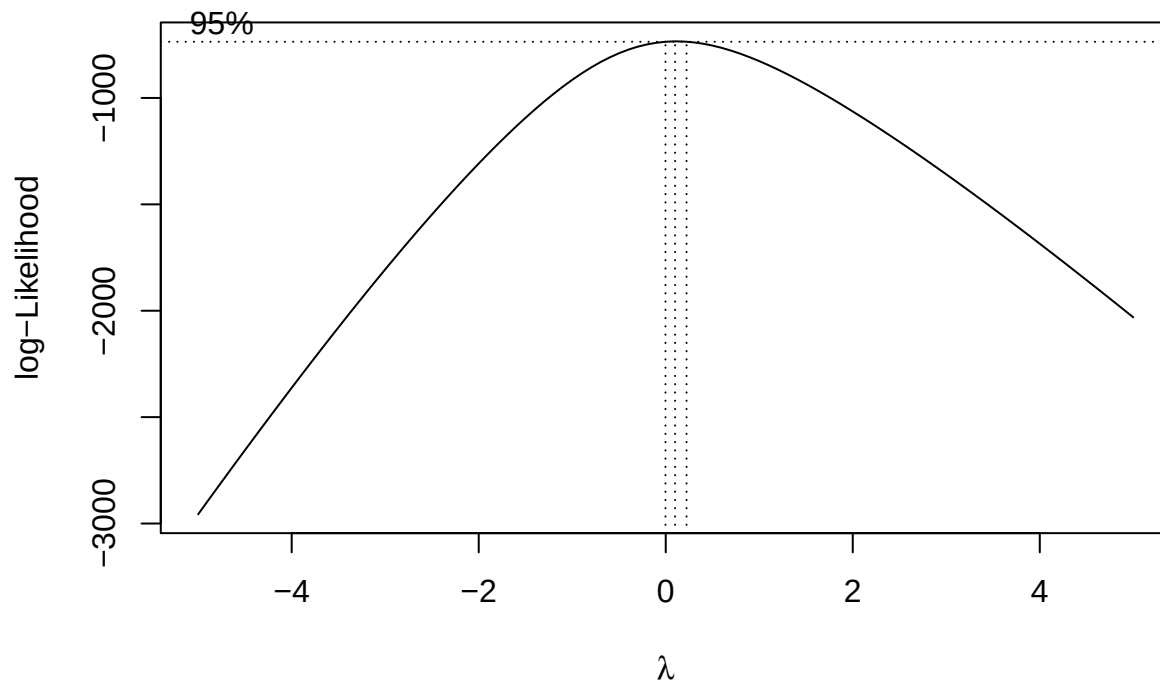
All variables are significant and R^2 is %74.

regression model assumptions

```
par(mfrow=c(2,2))
plot(lm.fit2)
```



```
par(mfrow=c(1,1))
library(MASS)
par(mfrow=c(1,1))
boxcox(lm.fit2,lambda=seq(-5,5,by=.1))
```



```
#getting the value of lambda
b=boxcox(lm.fit2,lambda=seq(-5,5,by=.1))
lambda<-b$x[which.max(b$y)]
lambda

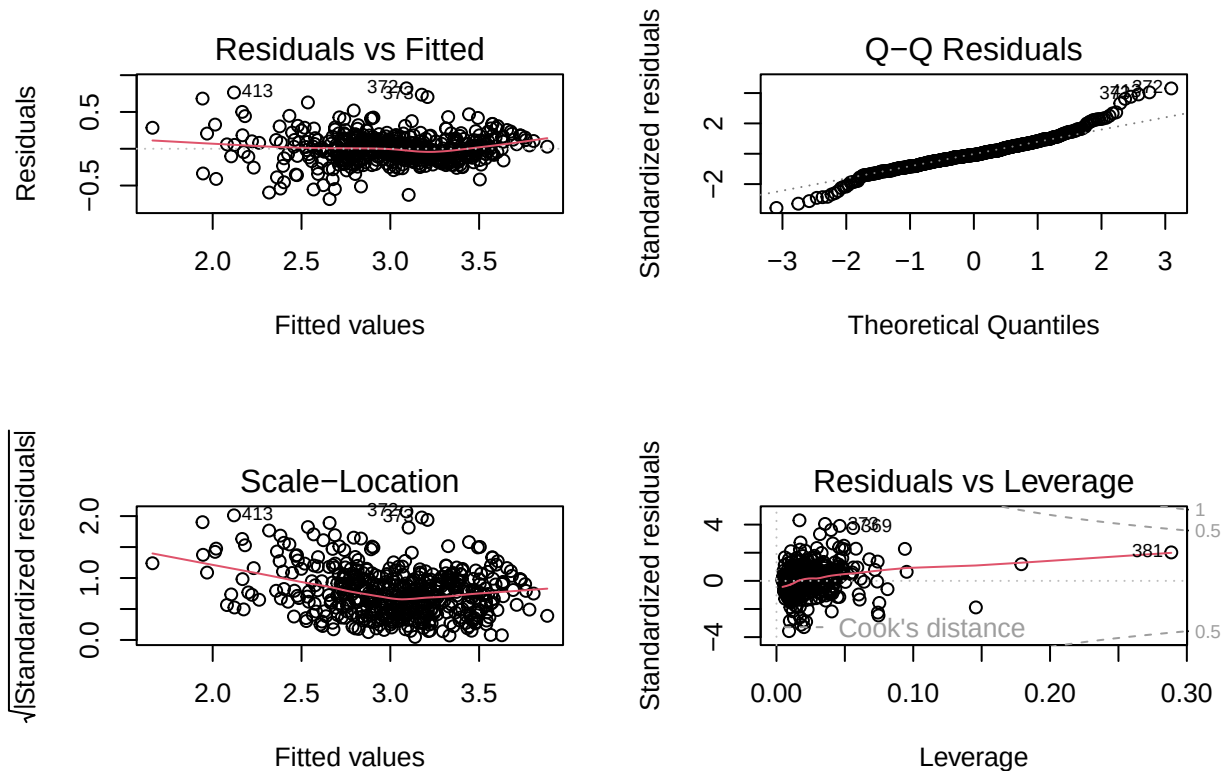
## [1] 0.1

lm.fit3<-lm(log(medv) ~ crim+zn+chas+nox+rm+dis+rad+tax+ptratio+lstat, data = Boston)
summary(lm.fit3)

##
## Call:
## lm(formula = log(medv) ~ crim + zn + chas + nox + rm + dis +
##     rad + tax + ptratio + lstat, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.68412 -0.10646 -0.01424  0.10064  0.82136
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.3107062  0.1970056  21.881  < 2e-16 ***
## crim        -0.0109074  0.0013226  -8.247  1.48e-15 ***
## zn           0.0011028  0.0005493   2.007  0.04525 *
## chas         0.1119520  0.0346574   3.230  0.00132 **
## nox          -0.7611199  0.1432456  -5.313  1.63e-07 ***
## rm           0.0849592  0.0164380   5.168  3.43e-07 ***
## dis          -0.0527383  0.0075405  -6.994  8.70e-12 ***
## rad           0.0127493  0.0025692   4.962  9.59e-07 ***
## tax          -0.0005807  0.0001369  -4.243  2.64e-05 ***
## ptratio      -0.0367346  0.0052402  -7.010  7.83e-12 ***
## lstat        -0.0296680  0.0019062 -15.564  < 2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1924 on 495 degrees of freedom
## Multiple R-squared:  0.7828, Adjusted R-squared:  0.7784
## F-statistic: 178.4 on 10 and 495 DF,  p-value: < 2.2e-16
```

```
par(mfrow=c(2,2))
plot(lm.fit3)
```



Box-Cox suggest log transformation distribution. QQ plot shows outliers. Graphs and Breusch Pagan test shows that the variances are unequal. The normality tests indicate that data is not normal.

```
##Breusch Pagan Test
```

H_o : Error variances are constant

H_a : Error variances are not constant

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## as.Date, as.Date.numeric
```

```
bptest(lm.fit2)
```

```
##
```

```
## studentized Breusch-Pagan test
```

```
##
```

```
## data: lm.fit2
```

```
## BP = 59.568, df = 10, p-value = 4.375e-09
```

```
#on transformed model
```

```
bptest(lm.fit3)
```

```
##
```

```
## studentized Breusch-Pagan test
```

```
##
```

```
## data: lm.fit3
```

```
## BP = 68.873, df = 10, p-value = 7.313e-11
```

Correlation Test for Normality

H_o : Errors are normally distributed

H_a : Errors are Not normally distributed

```
shapiro.test(lm.fit2$residuals)
```

```
##
```

```
## Shapiro-Wilk normality test
```

```
##
```

```
## data: lm.fit2$residuals
```

```
## W = 0.90821, p-value < 2.2e-16
```

Transforming the data

```
lm.fit3<- lm(log(medv) ~ crim + zn + chas + nox + rm + dis + rad + tax + ptratio + lstat, data = Boston)
```

```
summary(lm.fit3)
```

```
##
```

```
## Call:
```

```
## lm(formula = log(medv) ~ crim + zn + chas + nox + rm + dis +
```

```
##      rad + tax + ptratio + lstat, data = Boston)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -0.68412 -0.10646 -0.01424  0.10064  0.82136
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  4.3107062  0.1970056  21.881  < 2e-16 ***
```

```
## crim        -0.0109074  0.0013226  -8.247  1.48e-15 ***
```

```
## zn           0.0011028  0.0005493   2.007  0.04525 *
```

```
## chas         0.1119520  0.0346574   3.230  0.00132 **
```

```
## nox          -0.7611199  0.1432456  -5.313  1.63e-07 ***
```

```
## rm           0.0849592  0.0164380   5.168  3.43e-07 ***
```

```
## dis          -0.0527383  0.0075405  -6.994  8.70e-12 ***
```

```
## rad           0.0127493  0.0025692   4.962  9.59e-07 ***
```

```
## tax          -0.0005807  0.0001369  -4.243  2.64e-05 ***
```

```
## ptratio      -0.0367346  0.0052402  -7.010  7.83e-12 ***
```

```
## lstat        -0.0296680  0.0019062 -15.564  < 2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

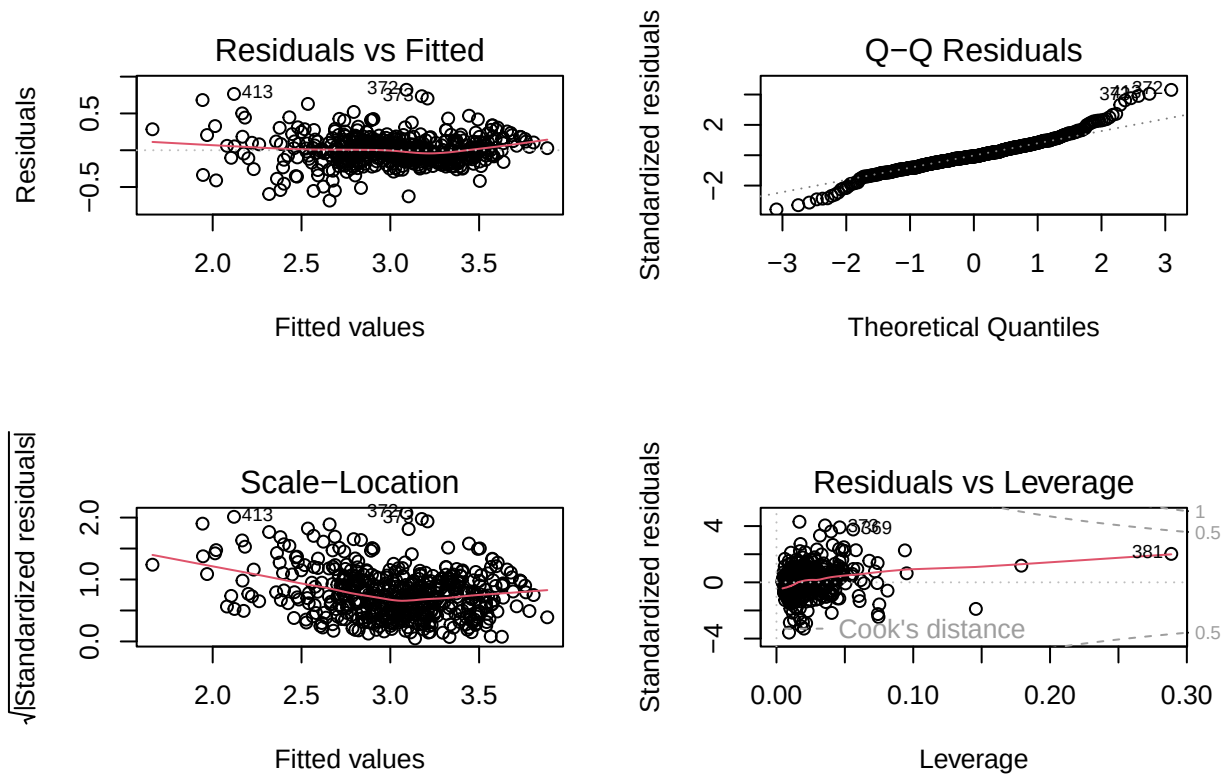
```
## Residual standard error: 0.1924 on 495 degrees of freedom
```

```
## Multiple R-squared:  0.7828, Adjusted R-squared:  0.7784
```



```
## F-statistic: 178.4 on 10 and 495 DF, p-value: < 2.2e-16
```

```
par(mfrow=c(2,2))
plot(lm.fit3)
```



```
bptest(lm.fit3)
```

```
##
## studentized Breusch-Pagan test
##
## data: lm.fit3
## BP = 68.873, df = 10, p-value = 7.313e-11
```

All variables are significant and R^2 is %79. QQ plot shows heavy tails with no outliers. Graphs and Breusch Pagan test shows that the variances are unequal.